

Dr. Romain Mottier

Computational Scientist — PhD in Applied Mathematics

📍 Basel, Switzerland ✉️ romain.mottier@outlook.com 📞 +33 6 23 23 24 29

🌐 romainmottier.github.io 🌐 github.com/romainmottier 🔗 linkedin.com/in/romain-mottier 🆔 0009-0001-2967-7694

French (native) • English (C2) • Valid Swiss work permit

PROFILE

Computational Scientist in Applied Mathematics & Physics. Six years building numerical methods for the simulation of complex physical problems: a strong background in **numerical analysis for PDEs**, strong programming skills in **C++**, **Python**, **Fortran** and **MATLAB**, and command of the underlying physics — solid and fluid mechanics, geophysics, wave propagation. I am also interested in the optimisation and portability on **high-performance computing (HPC) architectures**.

Position sought. Technical roles in R&D where rigorous mathematics and solid engineering create measurable value — and equally open to less hands-on positions, such as scientific project coordination or technical consulting.

I learn new fields fast and seek challenges in domains I do not yet know.

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher — Computational Mathematics

10/2025 – present

University of Basel, Dept. of Mathematics and Computer Science

Basel, Switzerland

- Design and implement (C++ template) **high-order solvers** for:
 1. wave propagation, using local time-stepping for efficient simulations.
 2. non-linear PDEs, with variable Sobolev exponents
- Authored 3 research articles (in preparation)

PhD Researcher — Applied Mathematics & Scientific Computing

10/2021 – 07/2025

*French Atomic Energy Commission (CEA) & École nationale des Ponts et Chaussées (ENPC)
Polytechnic Institute of Paris (IP Paris)*

Paris, France

- Developed and analysed **new high-order finite element methods** for coupled **elasto-acoustic wave propagation** — from mathematical analysis through to production code (C++ template – **Fortran 90**).
- Ran large-scale 2D/3D simulations on HPC cluster for geophysical wave-propagation studies, processing and post-processing.
- Authored **3 research articles** published in leading international journals and presented at **5 international conferences** in Europe and North America.
- Collaborated across three institutions (universities and CEA), reporting to both academic and industrial supervisors.

Research Intern — Computational Electromagnetics

03/2021 – 08/2021

ONERA, The French Aerospace Lab

Toulouse, France

- Computational discrete operator (CDO) schemes for the time-domain Maxwell equations, benchmarked against reference solutions.

Research Intern — Planetary Thermal Modelling

05/2020 – 08/2020

European Space Agency (ESA/ESTEC)

Noordwijk, Netherlands

- Implemented and solved a numerical heat-transfer model computing the surface and sub-surface temperature distribution on Mercury.

TECHNICAL SKILLS

Programming

C++ • Python • Fortran • MATLAB • Bash

Applied mathematics

Partial Differential Equations • Numerical Analysis • Linear Algebra • Optimisation

Scientific computing

High-Performance Computing (MPI, OpenMP) • Eigen • Intel MKL • BLAS/LAPACK

Numerical methods

Finite Element & Finite Volume Methods • Time Integration • Sparse Linear Solvers

Data & analysis

NumPy • Pandas • Matplotlib • Time-Series Analysis • Monte-Carlo • ParaView

AI-assisted engineering

LLM Coding Assistants (Claude Code, GitHub Copilot) • Prompt Engineering

Software engineering

Git • CMake • Linux & HPC Clusters • Code Review

Application domains

Wave Propagation & Fluid Mechanics • Geophysics • Electromagnetics

EDUCATION

PhD Researcher — Applied Mathematics & Scientific Computing <i>French Atomic Energy Commission (CEA) & École nationale des Ponts et Chaussées (ENPC)</i> <i>Polytechnic Institute of Paris (IP Paris)</i>	10/2021 – 07/2025 <i>Paris, France</i>
MSc Numerical Methods in Engineering (exchange year) <i>Universitat Politècnica de Catalunya (UPC)</i>	09/2020 – 02/2021 <i>Barcelona, Spain</i>
MSc in Engineering — Modelling and Fluid–Structure Computation <i>Université de Toulon, SeaTech Engineering School</i>	09/2018 – 09/2021 <i>Toulon, France</i>

TEACHING

Lecturer — Numerical Analysis of ODEs <i>University of Basel — final-year BSc course, Dept. of Mathematics and Computer Science</i>	Fall 2026 <i>Basel, Switzerland</i>
Supervision of Master’s semester projects <i>University of Basel — Numerical Methods for PDEs</i>	Fall 2025 – Spring 2026 <i>Basel, Switzerland</i>
• Supervision of Master’s students on finite element methods for linear elasticity (fall semester) and on the spectral element method (SEM) for the acoustic wave equation, adapted to industrial HPC architectures .	
Teaching Assistant — Numerical Methods <i>Paris Dauphine University — BSc Mathematics & Computer Science</i>	Spring 2023 <i>Paris, France</i>
• Lectures and lab sessions taught in Python (NumPy, Jupyter): nonlinear equations, polynomial interpolation, quadrature, direct and iterative linear solvers, eigenvalue computation.	
Teaching Assistant — Numerical Methods <i>Sorbonne University — MSc Computational Mechanics</i>	Fall 2022 <i>Paris, France</i>
• Lab sessions and numerical modelling projects: linear systems, finite differences, continuum mechanics.	

PUBLICATIONS & PREPRINTS

1. **Unfitted hybrid high-order methods stabilized by polynomial extension for elliptic interface problems.** *with E. Burman and A. Ern* — published in **SIAM J. Numer. Anal.** 64(3), 2026 — doi:10.1137/25M1741790
2. **Hybrid high-order methods for elasto-acoustic wave propagation in the time domain.** *with A. Ern, R. Khot and L. Guillot* — published in **ESAIM: M2AN** 59, 2685–2715, 2025 — doi:10.1051/m2an/2025062
3. **Elasto-acoustic wave propagation in geophysical media using hybrid high-order methods on general meshes.** *with A. Ern and L. Guillot* — published in **Comput. Methods Appl. Mech. Engrg.** 448, 118458, 2026 — doi:10.1016/j.cma.2025.118458
4. **Multilevel explicit local time-stepping for wave propagation with high-order hybrid nonconforming schemes (HDG/HHO).** *with M. J. Grote* — *in preparation*
5. **A hybrid high-order method for variable-exponent Leray–Lions problems.** *with A. Harnist* — *in preparation*

CONFERENCES

Talk	Conference on Analysis, Modeling and Numerics of Waves (Wave-NED)	Delft, NL — 08/2026
Talk	Computational Methods in Applied Mathematics (CMAM)	Vienna, AT — 07/2026
Talk	Mathematical and Numerical Aspects of Wave Propagation (WAVES)	Montreal, CA — 06/2026
Talk	World Congress on Computational Mechanics (WCCM)	Vancouver, CA — 07/2024
Talk	European Congress on Computational Methods in Engineering (ECCOMAS)	Lisbon, PT — 06/2024
Talk	National Congress of Numerical Analysis (CANUM)	Île de Ré, FR — 05/2024
Poster	American Geophysical Union Fall Meeting (AGU)	San Francisco, US — 12/2023
Poster	Congress of Young Researchers in Applied Mathematics (CJCMA)	Paris, FR — 09/2023

REFERENCES

Prof. Marcus J. Grote — Postdoctoral advisor
Professor, University of Basel
marcus.grote@unibas.ch

Prof. Alexandre Ern — PhD advisor
Senior Researcher, INRIA (SERENA)
alexandre.ern@enpc.fr

Dr. Laurent Guillot — Industrial PhD advisor
Researcher, CEA
laurent.guillot.blr@gmail.com

Dr. Guillaume Delay
Assistant Professor, Sorbonne University (LJLL)
guillaume.delay@sorbonne-universite.fr